Malhar Falke

Pune, India · +91 77381 59894 · falkemalhar388@gmail.com
linkedin.com/in/malhar-falke-660987291 · github.com/malharc373

Education

COEP Technological University (College of Engineering, Pune) Pune, India
B.Tech, Computer Engineering; CGPA 6.83/10 *Expected May 2027*
Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Artificial Intelligence, Embedded Systems.

Experience

Klimaforge Norway (Remote)
Machine Learning & Data Science Intern *May 2026 – Present*

- Built an above-ground forest biomass model for Estonia on ∼678k samples and 48 engineered features fused from Sentinel-1 SAR, Sentinel-2 optical and terrain data; XGBoost reached R^2 0.61 on a held-out split (cross-validated 0.58) and produced the national 2023 biomass map.
- Validated with 10 km spatial-block cross-validation (R^2 0.50) to expose optimism from spatial autocorrelation in random splits; caught and removed a canopy-height feature that leaked the target (R^2 1.00).
- Benchmarked the Prithvi-EO-2.0 geospatial foundation model (fine-tuning and embeddings) against engineered features; it underperformed at every training size, so the classical model shipped. Automated data ingestion with Google Earth Engine, CDSE and openEO.

COEP Technological University Pune, India
Embedded Systems Research Fellow *Jun 2025 – Aug 2025*

- Implemented SPI, I²C, UART, Bluetooth and Zigbee on PIC and ARM7/Cortex-M targets; wrote firmware for graphical and SPI TFT LCDs and PWM motor control.
- Evaluated FreeRTOS and μC/OS-II; authored self-learning modules and tested lab experiments for Embedded C and Embedded Linux.

Projects

AarthFit: MSME Financial Fitness Platform *Apr 2026 – Present*

- Built a multi-tenant SaaS for a business consultancy: Fastify + TypeScript REST API, React (Vite) SPA and PostgreSQL (TimescaleDB) with Knex migrations, in a pnpm monorepo.
- Implemented Redis-backed OTP/JWT authentication with role-based access control and audit logging; integrated Razorpay payments (webhook signature verification) and the Meta WhatsApp Cloud API for daily owner engagement.
- Designed the CFO Score engine (cashflow, compliance, profitability, balance sheet, governance) with PDF reports emailed via AWS SES; covered by Vitest integration tests against real Postgres and Redis.

IoT-IDS: Edge Network Intrusion Detection *2026 – Present*

- Built a flow-based IDS/IPS that detects 9 attack types across 4 categories; exported a 91.8 KB ONNX model with 8.1 μs single-flow inference on an Apple M4, plus a dependency-free C header for microcontrollers.
- Built a semantic feature-alignment framework over 11 public IDS datasets (CICIDS2017, UNSW-NB15, TON-IoT, Bot-IoT and others) to measure cross-dataset transfer; the study surfaced severe domain shift between datasets.

Leadership & Activities

Team Octane Racing Electric (Formula Student) COEP, Pune
E-Powertrain Engineer & Data Acquisition Lead *Oct 2023 – May 2026*

- Developed an STM32-based ECU, bringing CAN communication, DMA and an RTOS to the team's car for the first time in its history.
- Designed the data-acquisition system: STM32/Arduino logging of voltage, current, RPM, temperature and brake pressure with timestamped CAN-to-SD logging, Bluetooth live telemetry and Python analysis tools.
- Worked on battery integration, motor-controller interfacing and the low-voltage harness; competed at Formula Student Germany (Hockenheim, Aug 2025) and Formula Bharat (Jan 2024).

Skills

Languages: Python, C, C++, TypeScript/JavaScript, SQL, Java, Bash
Backend & Data: Node.js (Fastify), React, REST APIs, PostgreSQL, Redis, Docker, AWS (SES), pandas, XGBoost, PyTorch, ONNX
Embedded: STM32, ARM Cortex-M, PIC, FreeRTOS, CAN, SPI, I²C, UART
Tools: Git, Linux, Google Earth Engine, AI coding assistants (Claude Code, Codex) **Spoken:** English, Hindi, Marathi, German